

P2587R3

to_string or not to_string

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Audience:

LEWG

Project:

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Appendix A: Sample of code search results for `to_string`

References

"Though this be madness, yet there is method in 't." — Polonius

§ 1. Introduction

C++11 introduced a set of `std::to_string` overloads for integral and floating-point types. Fortunately for integral and unfortunately for floating-point overloads they are all defined in terms of `sprintf` inconsistently with C++ formatted output functions ([N4910]). Additionally, the choice of the floating-point format makes `std::to_string` of very limited use in practice. This paper proposes fixing these issues while retaining existing semantics of integral overloads.

§ 2. Revision history

Changes since R2:

- Added more poll results.

Changes since R1:

- Added poll results.
- Mentioned that the current proposal enables future extension to other types via `std::format`.
- Added an analysis of codesearch results to [§ 7 Impact on existing code](#).

Changes since R0:

- Added an entry to Annex C.
- Added a feature test macro.
- Applied the same changes to `to_wstring` in the wording.
- Added a reference to [STD_DISCUSSION] which proposed deprecating floating-point overloads of `std::to_string`.

§ 3. LEWG Poll (R2)

Poll: Forward P2587R2 for C++26 (Priority B2) to be confirmed by electronic polling.

SF	F	N	A	SA
----	---	---	---	----

11	6	2	0	0
----	---	---	---	---

Outcome: Strong consensus in favor

§ 4. LEWG Poll (R1)

Poll: Fix `to_string` as proposed by P2587R1 (`to_string` or not `to_string`).

SF	F	N	A	SA
4	9	3	2	1

Outcome: Weak consensus

§ 5. Examples

Consider the following example:

```
auto loc = std::locale("uk_UA.UTF-8");
std::locale::global(loc);
std::cout.imbue(loc);
setlocale(LC_ALL, "C");

std::cout << "iostreams:\n";
std::cout << 1234 << "\n";
std::cout << 1234.5 << "\n";

std::cout << "\nto_string:\n";
std::cout << std::to_string(1234) << "\n";
std::cout << std::to_string(1234.5) << "\n";

setlocale(LC_ALL, "uk_UA.UTF-8");

std::cout << "\nto_string (uk_UA.UTF-8 C locale):\n";
std::cout << std::to_string(1234) << "\n";
std::cout << std::to_string(1234.5) << "\n";
```

It prints:

```
iostreams:
1 234
1 234,5

to_string:
1234
1234.500000

to_string (uk_UA.UTF-8 C locale):
1234
1234,500000
```

Since `std::to_string` uses the global C locale and no grouping the integral overloads are effectively unlocalized. The output of floating-point overloads is inconsistent with that of `iostreams` because the former takes the decimal point from the global C locale and doesn't do grouping.

Additionally, due to an unfortunate choice of the fixed format in the floating-point overloads they are only useful for numbers in a limited exponent range. For example:

```
std::cout << std::to_string(std::numeric_limits<double>::max());
```

prints

```
179769313486231570814527423731704356798070567525844996598917476803157260786
387605895586327668781715404589535143824642343213268894641827684675467035375
604991057655128207624549009038932894407586850845513394230458323690322294816
559332123348274797826204144723168738177180919299881250404026184124858368.00
```

(line breaks inserted for readability)

Here only the first 17 digits are meaningful, the next 292 are so-called "garbage" digits ([\[DRAGON\]](#)). And finally we have 6 meaningless zeros after a possibly localized decimal point.

Formatting of small floating-point numbers is even less useful. For example:

```
std::cout << std::to_string(-1e-7);
```

prints

```
-0.000000
```

In fact almost half of floating-point numbers are formatted as zero and there is a precision loss or garbage digit output in many other cases.

Unfortunately issues with floating-point overloads of `std::to_string` have not been discussed in the paper that proposed this facility ([N1803], [N2408]). However, they were brought up as early as 2015 in a mailing list discussion ([STD_DISCUSSION]) which proposed deprecating floating-point overloads of `std::to_string` but no paper came out of that.

§ 6. Proposal

Redefine `std::to_string` in terms of `std::format` which in turn uses `std::to_chars` making more explicit the fact that integral overloads are unlocalized and changing the format of floating-point overloads to also be unlocalized and use the shortest decimal representation.

The following table shows the changes in output for the following code:

```
setlocale(LC_ALL, "C");  
auto output = std::to_string(input);
```

input	output	
	before	after
42	42	42
0.42	0.420000	0.42
-1e-7	-0.000000	-1e-7
1.7976931348623157e+308	1797693134862315708145274 2373170435679807056752584 4996598917476803157260780 0285387605895586327668781 7154045895351438246423432 1326889464182768467546703 5375169860499105765512820 7624549009038932894407586 8508455133942304583236903 2229481658085593321233482	1.7976931348623157e+308

input	output	
	before	after
	7479782620414472316873817	
	7180919299881250404026184	
	124858368.000000	

and similarly with the global C locale set:

```
setlocale(LC_ALL, "uk_UA.UTF-8");
auto output = std::to_string(input);
```

input	output	
	before	after
12345	12345	12345
1234.5	1234,500000	1234.5

In addition to fixing the broken behavior for floating point, this proposal enables an easy and natural extension of `std::to_string` to other types via `std::format` (not proposed in this paper).

§ 7. Impact on existing code

This change will affect the output of `std::to_string` with floating-point arguments. In most cases it will result in a more precise and/or shorter output. In cases where the C locale is explicitly set the decimal point will no longer be localized.

In order to estimate the practical impact on existing code we looked at [\[CODESEARCH\]](#) results for `std::to_string`. 876 million lines of code contained only ~11 thousand calls to this function. For comparison, there are ~20 thousand uses of `std::stringstream` and ~390 thousand calls to `sprintf`. We further analyzed a random sample of 100 `std::to_string` invocations listed in [Appendix A](#). 91 of them are called with integral arguments and only 7 contain calls with floating-point arguments. Note that some of these 7 results contain calls with integral and FP arguments on the same line. The two remaining cases are a generic wrapper around `std::to_string` and an overload for a user-defined type.

Three of the seven FP calls are in test code with one of them used to format zero:

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-4/src/tests/unit-tests/signal-processing-blocks/tracking/gps_l1_ca_dll_pll_tracking_test.cc:916](#)

```
g5.disable  
}  
g5.set_title("Code  
g5.set_grid();  
g5.set_xlabel("Time
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.9-5/src/tests/system-tests/obs_gps_l1_system_test.cc:229](#)

```
config->set_property("SignalSource.sampling_frequency", std::to_str  
config->set_property("SignalSource.item_type", "ibyte");  
config->set_property("SignalSource.samples", std::to_string(zero));  
  
// Set the Signal Conditioner
```

[actcd19/main/t/trilinos/trilinos_12.12.1-7/packages/stk/stk_simd/unit_tests/SimdFixture.hpp:268](#)

```
virtual std::string get_input_vals(int i) const override {  
    return std::to_string(x.scalar(i))+" "+std::to_string(y.scalar(i));  
}  
};
```

Two of the calls are used for informational messages (writing a comment to a file and printing status):

[actcd19/main/p/pdal/pdal_1.7.2-2/vendor/kazhdan/SurfaceTrimmerMain.cpp:127](#)

```
if (Trim.set)  
    comments.push_back(leader + Trim.name + " " +  
        std::to_string(Trim.value));  
if (Smooth.set)  
    comments.push_back(leader + IslandAreaRatio.name + " " +
```

[actcd19/main/m/madness/madness_0.10.1~gite4aa500e-3/src/apps/chem/CC2.cc:131](#)

```
bool doubles_converged = true;
for(size_t iter=0; iter < parameters.iter_max_6D; iter++){
    output_subsection("Macroiteration " + std::to_string(iter) + " with
    CC_Timer timer_iter_all(world,"Macroiteration " + stringify(iter))
```

In one case `std::to_string` is used to write an FP value to a config file. It is likely a latent bug because writing a config in one locale may render it unreadable in a different one unless special measures are taken to fix the locale.

[actcd19/main/p/performous/performous_1.1+git20181118-2/game/configuration.cc:310](#)

```
else if (type == "int") entryNode->set_attribute("value", st
else if (type == "bool") entryNode->set_attribute("value",
else if (type == "float") entryNode->set_attribute("value",
else if (item.get_type() == "string") xmlpp::add_child_ele
else if (item.get_type() == "string_list") {
```

The last call is used to generate a human-readable name for some object, possibly for debug purposes. We haven't found any uses of the enclosing function though.

[actcd19/main/p/psi4/psi4_1.2.1-2/psi4/src/psi4/libmints/cdsalclist.cc:312](#)

```
name += com.coef > 0.0 ? "+" : "-";
name += std::to_string(std::fabs(com.coef)) + " ";
name += molecule_>label(com.atom);
if (com.xyz == 0)
```

In summary, the number of uses of `to_string` with floating point arguments turned out to be very low and the cases that we analyzed are mostly non-critical such as tests and debugging/logging use cases. Some of them may even contain latent bugs because the use of the locale is unintentional.

As a mitigation mechanism we could provide `to_string_localized` (not currently proposed) with the old behavior which would provide a simple migration path.

§ 8. Implementation

{fmt} implements proposed changes in `fmt::to_string`.

§ 9. Wording

Add an entry for `__cpp_lib_to_string` to section "Header `<version>` synopsis [[version.syn](#)]", in a place that respects the table's current alphabetic order:

```
#define __cpp_lib_to_string **placeholder** // also in <string>
```

Modify subsection "Numeric conversions [[string.conversions](#)]":

```
string to_string(int val);
string to_string(unsigned val);
string to_string(long val);
string to_string(unsigned long val);
string to_string(long long val);
string to_string(unsigned long long val);
string to_string(float val);
string to_string(double val);
string to_string(long double val);
```

~~7 Returns: Each function returns a string object holding the character representation of the value of its argument that would be generated by calling `sprintf(buf, fmt, val)` with a format specifier of `"%d"`, `"%u"`, `"%ld"`, `"%lu"`, `"%lld"`, `"%llu"`, `"%f"`, `"%F"`, or `"%Lf"`, respectively, where `buf` designates an internal character buffer of sufficient size.~~

7 Returns: `format("{} ", val)`.

```
wstring to_wstring(int val);
wstring to_wstring(unsigned val);
wstring to_wstring(long val);
wstring to_wstring(unsigned long val);
wstring to_wstring(long long val);
wstring to_wstring(unsigned long long val);
wstring to_wstring(float val);
wstring to_wstring(double val);
wstring to_wstring(long double val);
```

~~14 Returns: Each function returns a `wstring` object holding the character representation of the value of its argument that would be generated by calling `swprintf(buf, buffsz, fmt, val)` with a format specifier of `L"%d"`, `L"%u"`, `L"%ld"`, `L"%lu"`, `L"%lld"`, `L"%llu"`, `L"%f"`, `L"%f"`, or `L"%Lf"`, respectively, where `buf` designates an internal character buffer of sufficient size `buffersz`.~~

14 Returns: `format(L"{ }", val)`.

Add to Annex C (informative) Compatibility [\[diff\]](#) the following new subclause:

C.? C++ and ISO C++ 2026 [\[diff.cpp23\]](#)

This subclause lists the differences between C++ and ISO C++ 2023 (ISO/IEC 14882:2023, Programming Languages — C++), by the chapters of this document.

C.?.1 Clause 23: strings library [\[diff.cpp23.strings\]](#)

Affected subclauses: 23.4

Change: Output of floating-point overloads of `to_string` and `to_wstring`.

Rationale: Prevent loss of information and improve consistency with other formatting facilities.

Effect on original feature: `to_string` and `to_wstring` function calls that take floating-point arguments may produce a different output. For example:

```
auto s = std::to_string(1e-7); // "1e-7"
// previously "0.000000" with '.' possibly
// changed according to the global C locale
```

§ Appendix A: Sample of code search results for `to_string`

Searching for `std::to_string` ...

2489599 source files searched.

11408 matches found.

Here is a random sample of matches...

[actcd19/main/u/ultracopier/ultracopier_1.6.1.3-1/plugins/Themes/Oxygen/ThemesFactory.cpp:339](#)

```

    if(!ui->checkBoxShowSpeed->isChecked())
        return;
    ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice,"value: "+s
    switch(value)
    {

```

[actcd19/main/r/re2/re2_20190101+dfsg-2/re2/testing/re2_test.cc:870](#)

```

    ASSERT_TRUE(RE2::FullMatch("-100", "( -?\\d+)", &v2)); ASSERT_EQ(v2,
    str = std::to_string(max);
    ASSERT_TRUE(RE2::FullMatch(str, "( -?\\d+)", &v)); ASSERT_EQ(v, n

```

[actcd19/main/s/scilab/scilab_6.0.1-7/modules/hdf5/src/cpp/handle_properties.cpp:2683](#)

```

        dims[0] = 1;
        dims[1] = static_cast<int>(path.size());
        writeIntegerMatrix6(node, std::to_string(i).data(), H5T_NATIVE
    }
}

```

[actcd19/main/r/rosegarden/rosegarden_18.12-1/src/sound/MidiEvent.cpp:181](#)

```

        sharpflat = std::to_string(-tonality) + " flat";
    } else {
        sharpflat = std::to_string(tonality);
        sharpflat += " sharp";
    }

```

[actcd19/main/n/nheko/nheko_0.6.3-1/mtxclient/lib/http/client.cpp:137](#)

```
{  
    std::map<std::string, std::string> params;  
    params.emplace("limit", std::to_string(limit));  
  
    get<mtx::responses::Notifications>(
```

[actcd19/main/u/ultracopier/ultracopier_1.6.1.3-1/plugins/CopyEngine/Ultracopier/WriteThread.cpp:72](#)

```
    if(stopIt)  
    {  
        ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice, "[" + std::to_string(stopIt) + "]"  
        emit closed();  
        return false;  
    }
```

[actcd19/main/p/pdns/pdns_4.1.6-1/pdns/dnsname.cc:125](#)

```
    if(newpos < offset) {  
        if(newpos < (int) minOffset)  
            throw std::range_error("Invalid label position during decompression");  
        if (++depth > 100)  
            throw std::range_error("Abort label decompression after 100 recursive calls");  
    }
```

[actcd19/main/c/chromium/chromium_72.0.3626.121-1/chrome/browser/ui/search/local_ntp_browsertest.cc:637](#)

```
    for (int i = kDefaultMostVisitedItemCount; i < kDefaultCustomLinkMaxCount; ++i) {  
        std::string rid = std::to_string(i + 100);  
        std::string url = "https://" + rid + ".com";  
        std::string title = "url for " + rid;  
    }
```

[actcd19/main/s/simgrid/simgrid_3.21+dfsg-4/docs/source/tuto_s4u/master-workers-lab4.cpp:62](#)

```
/* - Select a worker in a round-robin way */
aid_t worker_pid          = actors.at(task_id % actors.size(
std::string mailbox_name  = std::string("worker-") + std::to_s
simgrid::s4u::MailboxPtr mailbox = simgrid::s4u::Mailbox::by_name(n
```

[actcd19/main/s/sysdig/sysdig_0.22.0-1/userspace/libinsp/mesos.cpp:618](#)

```
if(m_inacti
{
    g_l
}
```

[actcd19/main/d/dnsdist/dnsdist_1.3.3-2/dns.cc:88](#)

```
if(offset < d_length)
    return d_ptr[offset];
throw runtime_error("out of bounds: "+std::to_string(offset)+" >= '
}
private:
```

[actcd19/main/c/chromium/chromium_72.0.3626.121-1/extensions/browser/api/display_source/display_source_apitestbase.cc:487](#)

```
net_result = socket_>Bind(local_point);
if (net_result == net::OK) {
    udp_port_ = std::to_string(port);
    // When we got an udp socket established and udp port is known
    // Change sink's status to connected and proceed with the test.
```

[actcd19/main/p/pdns/pdns_4.1.6-1/pdns/validate.cc:1054](#)

```
if (!DNSCryptoKeyEngine::isDigestSupported(ds.d_digesttype)) {  
    LOG("Discarding DS "<<ds.d_tag<<" because we don't support digest r  
    return false;  
}
```

[actcd19/main/d/dnsdist/dnsdist_1.3.3-2/fstrm_logger.cc:24](#)

```
res = fstrm_writer_options_add_content_type(d_fwopt, DNSTAP_CONTENT  
if (res != fstrm_res_success) {  
    throw std::runtime_error("FrameStreamLogger: fstrm_writer_options  
}
```

[actcd19/main/p/pdal/pdal_1.7.2-2/vendor/kazhdan/SurfaceTrimmerMain.cpp:127](#)

```
if (Trim.set)  
    comments.push_back(leader + Trim.name + " " +  
        std::to_string(Trim.value));  
if (Smooth.set)  
    comments.push_back(leader + IslandAreaRatio.name + " " +
```

[actcd19/main/p/pybind11/pybind11_2.2.4-2/tests/test_kwargs_and_defaults.cpp:15](#)

```
TEST_SUBMODULE(kwargs_and_defaults, m) {  
    auto kw_func = [](int x, int y) { return "x=" + std::to_string(x) +  
  
    // test_named_arguments
```

[actcd19/main/p/pdns-recursor/pdns-recursor_4.1.11-1/dnsparser.cc:652](#)

```

        if(d_notyouroffset > d_length)
            throw std::out_of_range("dns packet out of range: "+std::to_string(d_notyouroffset)
            + std::to_string(d_length) );
    }
    void rewindOffset(uint16_t by)

```

[actcd19/main/c/chromium/chromium_72.0.3626.121-1/third_party/webrtc/rtc_base/string_to_number_unittest.cc:47](#)

```

constexpr T zero_value = 0;
const std::string min_string = std::to_string(min_value);
const std::string max_string = std::to_string(max_value);
EXPECT_EQ(min_value, StringToNumber<T>(min_string));
EXPECT_EQ(min_value, StringToNumber<T>(min_string.c_str()));

```

[actcd19/main/c/chromium/chromium_72.0.3626.121-1/pc/disable/tracing.patch/content/browser/browser_child_process_host_impl.cc:415](#)

```

std::string log_message = "Bad message received of type: ";
if (message.IsValid()) {
    log_message += std::to_string(message.type());
} else {
    log_message += "unknown";
}

```

[actcd19/main/q/qtwebengine-opensource-src/qtwebengine-opensource-src_5.11.3+dfsg-2/src/3rdparty/chromium/third_party/swiftshader/third_party/subzero/src/IceInst.cpp:603](#)

```

        Ctx, ".L" + FuncName.toString() + "$jumptable$__" + std::to_string(FuncName.getID())
return GlobalString::createWithString(
    Ctx, "$J" + std::to_string(FuncName.getID()) + "_" + std::to_string(FuncName.getID())
}
} // end of anonymous namespace

```

[actcd19/main/r/rocksdb/rocksdb_5.17.2-3/examples/compact_files_example.cc:155](#)

```
// because of options.level0_stop_writes_trigger
for (int i = 1000; i < 99999; ++i) {
    db->Put(WriteOptions(), std::to_string(i),
           std::string(500, 'a' + (i % 26)));
}
```

[actcd19/main/t/trilinos/trilinos_12.12.1-](#)

[7/packages/seacas/libraries/ioss/src/xdmf/Ioxf_DatabaseIO.C:1054](#)

```
std::string FinalName(hdfname.tailname().c_str());
FinalName += "://" + block_name + "/Iteration " + std::to_string(i);
XdmfArray *ScalarArray = new XdmfArray();
if (sizeof(double) == 8)
```

[actcd19/main/q/qtwebengine-opensource-src/qtwebengine-opensource-src_5.11.3+dfsg-](#)

[2/src/3rdparty/chromium/third_party/swiftshader/src/OpenGL/compiler/preprocessor/Macro.cpp:40](#)

```
Token token;
token.type = Token::CONST_INT;
token.text = std::to_string(value);

std::shared_ptr<Macro> macro = std::make_shared<Macro>();
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-4/src/tests/unit-tests/signal-processing-](#)

[blocks/tracking/gps_l1_ca_dll_pll_tracking_test.cc:916](#)

```
g5.disable()
}
g5.set_title("Code
g5.set_grid();
g5.set_xlabel("Time
```

[actcd19/main/a/antlr4-cpp-runtime/antlr4-cpp-runtime_4.7.2+dfsg-1/runtime/src/atn/ATNSerializer.cpp:475](#)

```
size_t s = data[p++];  
buf.append("mode ")  
.append(std::to_string(i))  
.append(":")  
.append(std::to_string(s))
```

[actcd19/main/q/qtwebengine-opensource-src/qtwebengine-opensource-src_5.11.3+dfsg-2/src/3rdparty/chromium/third_party/peretto/src/protozero/protoc_plugin/protozero_generator.cc:322](#)

```
stub_h_>Print("$name$ = $number$,\\n", "name",  
              value_name_prefix + value->name(), "number",  
              std::to_string(value->number()));  
}
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-4/src/algorithms/tracking/gnuradio_blocks/glonass_l2_ca_dll_pll_c_aid_tracking_sc.cc:559](#)

```
try  
{  
    d_dump_filename.append(std::to_string(d_cha  
    d_dump_filename.append(".dat");  
    d_dump_file.exceptions(std::ifstream::failt
```

[actcd19/main/u/ultracopier/ultracopier_1.6.1.3-1/plugins/CopyEngine/Ultracopier/ReadThread.cpp:623](#)

```
bool ReadThread::internalReopen()  
{  
    ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice, "["+std::to  
    stopIt=false;
```

```
if(file.isOpen())
```

[actcd19/main/c/chromium/chromium_72.0.3626.121-](#)

[1/third_party/webrtc/modules/video_coding/codecs/test/videocodec_test_mediacodec.cc:137](#)

```
const int width = std::get<0>(resolution);  
const int height = std::get<1>(resolution);  
config.filename = std::string("foreman_") + std::to_string(width)  
                + std::to_string(height);  
config.filepath = ResourcePath(config.filename, "yuv");
```

[actcd19/main/b/bedops/bedops_2.4.35+dfsg-1/interfaces/general-headers/data/bed/Bed.hpp:389](#)

```
static const std::string tab = "\t";  
return std::string(chrom_) + tab + std::to_string(start_) + tab +  
        + std::string(id_);  
}
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-4/src/core/receiver/gnss_block_factory.cc:189](#)

```
if (ID != -1)  
{  
    role = "SignalSource" + std::to_string(ID);  
}  
}
```

[actcd19/main/m/mariadb-10.3/mariadb-10.3_10.3.13-1/storage/rocksdb/rdb_i_s.cc:498](#)

```
{"TARGET_FILE_SIZE_BASE", std::to_string(opts.target_file_size_  
{"TARGET_FILE_SIZE_MULTIPLIER",  
std::to_string(opts.target_file_size_multiplier)},  
{"MAX_BYTES_FOR_LEVEL_BASE",
```

```
std::to_string(opts.max_bytes_for_level_base)}},
```

[actcd19/main/c/cmake/cmake_3.12.3-3/Source/cmStateSnapshot.cxx:338](#)

```
std::to_string(cmVersion::GetMajorVersion()));  
this->SetDefinition("CMAKE_MINOR_VERSION",  
std::to_string(cmVersion::GetMinorVersion()));  
this->SetDefinition("CMAKE_PATCH_VERSION",  
std::to_string(cmVersion::GetPatchVersion()));
```

[actcd19/main/p/pdns/pdns_4.1.6-1/pdns/pkcs11signers.cc:234](#)

```
    } else {  
        logError("C_GetTokenInfo");  
        throw PDNSException("Cannot get token info for slot " + std::to_string(slot));  
    }  
}
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-](#)

[4/src/algorithms/telemetry_decoder/gnuradio_blocks/gps_l2c_telemetry_decoder_cc.cc:112](#)

```
{  
    d_dump_filename = "telemetry_L2CM_";  
    d_dump_filename.append(std::to_string(d_channel));  
    d_dump_filename.append(".dat");  
    d_dump_file.exceptions(std::ifstream::failbit | std::ifstream::append | std::ifstream::trunc);
```

[actcd19/main/s/scilab/scilab_6.0.1-7/modules/hdf5/src/cpp/handle_properties.cpp:984](#)

```
int col = 0;  
int* path = nullptr;  
getHandleIntVector(node, std::to_string(i).data(), &row, &col,
```

```
std::vector<int> res(row * col);  
for (int j = 0; j < row * col; ++j)
```

[actcd19/main/p/pdal/pdal_1.7.2-2/vendor/arbiter/arbiter.cpp:1447](#)

```
if (!headers.count("Content-Length"))  
{  
    headers["Content-Length"] = std::to_string(data.size());  
}  
return m_pool.acquire().post(typedPath(path), data, headers, query)
```

[actcd19/main/f/fw4spl/fw4spl_17.2.0-2/SrcLib/core/fwCommand/test/tu/src/UndoRedoManagerTest.cpp:229](#)

```
{  
    BogusCommand::sptr testCmdI =  
        std::make_shared<BogusCommand>(BogusCommand("testCmd" + std::to_string(i)));  
    undoRedoManager.enqueue(testCmdI);  
}
```

[actcd19/main/s/sysdig/sysdig_0.22.0-1/userspace/libsinsp/socket_handler.h:1229](#)

```
si  
g_logger.log("Socket handle  
"  
"  
si
```

[actcd19/main/p/performous/performous_1.1+git20181118-2/game/configuration.cc:310](#)

```
else if (type == "int") entryNode->set_attribute("value", st  
else if (type == "bool") entryNode->set_attribute("value",  
else if (type == "float") entryNode->set_attribute("value",
```

```
else if (item.get_type() == "string") xmlpp::add_child_element
else if (item.get_type() == "string_list") {
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-4/src/core/receiver/gnss_flowgraph.cc:1238](#)

```
try
{
    sat_ = configuration_>property("Channel" +
}
catch (const std::exception& e)
```

[actcd19/main/q/qtwebengine-opensource-src/qtwebengine-opensource-src_5.11.3+dfsg-2/src/3rdparty/chromium/services/ui/ws/window_tree_unittest.cc:1539](#)

```
// would be reset to 0 before sending back to clients.
EXPECT_EQ("OnCaptureChanged new_window=0," +
    std::to_string(kEmbedTreeWindowId) +
    " old_window=" + kNextWindowClientIdString + "," +
    std::to_string(embed_child_window_id.sink_id()),
```

[actcd19/main/g/gazebo/gazebo_9.6.0-1/gazebo/physics/dart/DARTScrewJoint.cc:95](#)

```
{
    return this->dataPtr->GetCached<ignition::math::Vector3d>(
        "Axis" + std::to_string(_index));
}
```

[actcd19/main/q/qtcreator/qtcreator_4.8.2-1/src/libs/3rdparty/botan/src/lib/asn1/asn1_obj.cpp:48](#)

```
else
{
    msg << std::to_string(type_tag);
```

```
}
```

[actcd19/main/l/llvm-toolchain-7/llvm-toolchain-7_7.0.1-8/.pc/kfreebsd/kfreebsd-triple-clang.diff/clang/lib/Driver/ToolChains/Clang.cpp:3358](#)

```
if (FunctionAlignment) {  
    CmdArgs.push_back("-function-alignment");  
    CmdArgs.push_back(Args.MakeArgString(std::to_string(FunctionAlignme  
}
```

[actcd19/non-free/i/intel-media-driver-non-free/intel-media-driver-non-free_18.4.0+ds1-1/media_driver/agnostic/common/os/mos_os.c:639](#)

```
auto &cmdResInfoPtrs = GetCmdResPtrs(p0sInterface);  
  
outputFile << "--PerfTag: " << std::to_string(p0sInterface->pfnGetF  
    << cmdResInfoPtrs.size() << " --Dump Count: " << ++m_cnt << enc
```

[actcd19/main/o/osm2pgsql/osm2pgsql_0.96.0+ds-2/middle-pgsql.cpp:267](#)

```
};  
  
copy_buffer = std::to_string(node.id());  
copy_buffer += delim;
```

[actcd19/main/l/lizardfs/lizardfs_3.12.0+dfsg-3/src/master/chunk_goal_counters.cc:63](#)

```
if (it == counters_.end() || it->goal != goal) {  
    throw ChunkGoalCounters::InvalidOperation(  
        "Trying to remove non-existent goal: " + st
```

```
}  
it->count--;
```

[actcd19/main/c/cbmc/cbmc_5.10-5/src/solvers/smt2/smt2_conv.cpp:802](#)

```
else if(type.id()==ID_unsignedbv)  
{  
    return "u"+std::to_string(to_unsignedbv_type(type).get_width());  
}  
else if(type.id()==ID_c_bool)
```

[actcd19/main/p/pcsx2/pcsx2_1.5.0~gfc1d9aef0+dfsg-1/plugins/cdvdGigahertz/src/CDVD.cpp:51](#)

```
#ifndef PCSX2_DEBUG  
static std::string s_libname("cdvdGigahertz " + std::to_string(SVN_REV))  
#else  
static std::string s_libname("cdvdGigahertz Debug " + std::to_string(SVN_REV))
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.9-5/src/tests/system-tests/obs_gps_l1_system_test.cc:229](#)

```
config->set_property("SignalSource.sampling_frequency", std::to_string(sampling_frequency));  
config->set_property("SignalSource.item_type", "ibyte");  
config->set_property("SignalSource.samples", std::to_string(zero));  
  
// Set the Signal Conditioner
```

[actcd19/main/c/cbmc/cbmc_5.6-1/src/cegis/jsa/genetic/jsa_source_provider.cpp:55](#)

```
result+=std::to_string(__CPROVER_JSA_MAX_QUERY_SIZE);  
result+="\n#define __CPROVER_JSA_MAX_PRED_SIZE ";  
result+=std::to_string(__CPROVER_JSA_MAX_PRED_SIZE);
```

```
result+="\n#define __CPROVER_JSA_NUM_PRED_OPS ";
result+=std::to_string(__CPROVER_JSA_NUM_PRED_OPS);
```

[actcd19/main/p/pcsx2/pcsx2_1.5.0~gfc1d9aef0+dfsg-1/common/src/Utilities/Perf.cpp:141](#)

```
    ml.method_name = m_prefix;
#else
    std::string name = std::string(m_prefix) + "_" + std::to_string(pc)
    ml.method_id = iJIT_GetNewMethodID();
    ml.method_name = (char *)name.c_str();
```

[actcd19/main/u/ultracopier/ultracopier_1.6.1.3-1/plugins/CopyEngine/Ultracopier/TransferThread.cpp:1808](#)

```
    ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice, "["+std::to_string(transfer_stat)+"]");
    ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice, "["+std::to_string(transfer_stat)+"]");
    ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice, "["+std::to_string(transfer_stat)+"]");
    switch(transfer_stat)
    {
```

[actcd19/main/m/modsecurity/modsecurity_3.0.3-1/src/operators/verify_ssn.cc:80](#)

```
    str_area.append(std::to_string(num[0]) + std::to_string(num[1]) +
        std::to_string(num[2]));

    str_grp.append(std::to_string(num[3]) + std::to_string(num[4]));
```

[actcd19/main/m/monero/monero_0.13.0.4-2/src/device/device_ledger.cpp:1234](#)

```
    this->controle_device->mlsag_prehash(blob_x, inputs_size_x, outputs_size_x);
    if (inputs_size) {
        log_message("mlsag_prehash", (std::string("inputs_size not null") + std::to_string(inputs_size_x) + " outputs_size not null" + std::to_string(outputs_size_x)));
```

```
}  
this->key_map.log();
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-](#)

[4/src/algorithms/acquisition/adapters/galileo_e1_pcps_cccwsr_ambiguous_acquisition.cc:225](#)

```
{  
    bool cboc = configuration_->property(  
        "Acquisition" + std::to_string(channel_) + ".cboc", fal  
  
    char signal[3];
```

[actcd19/main/m/madness/madness_0.10.1~gite4aa500e-3/src/apps/chem/CC2.cc:131](#)

```
bool doubles_converged = true;  
for(size_t iter=0; iter < parameters.iter_max_6D; iter++){  
    output_subsection("Macroiteration " + std::to_string(iter) + " wi  
    CC_Timer timer_iter_all(world,"Macroiteration " + stringify(iter)
```

[actcd19/main/s/simgrid/simgrid_3.21+dfsg-4/src/instr/instr_private.hpp:162](#)

```
    return stream.str();  
}  
std::string display_size() override { return std::to_string(send_size  
};
```

[actcd19/main/h/horizon-eda/horizon-eda_0.20181108-1/src/gen-pkg/gen-pkg.cpp:163](#)

```
auto &pad = pkg.pads.emplace(std::piecewise_construct, std::for  
    .first->second;  
pad.name = std::to_string(i + 1);
```

```
pad.placement.shift.y = y0 - i * pitch;  
pad.parameter_set[ParameterID::HOLE_DIAMETER] = 1_mm;
```

[actcd19/main/s/sysdig/sysdig_0.22.0-1/userspace/libsinsp/socket_handler.h:1522](#)

```
if(pending_reqs)  
{  
    g_logger.log("Socket handler: number of pending DNS  
                (pending_reqs > 10) ? sinsp  
                )  
}
```

[actcd19/main/libf/libfreesrp/libfreesrp_0.3.0-2/src/freesrp_impl.cpp:85](#)

```
libusb_close(_freesrp_handle);  
_freesrp_handle = nullptr;  
throw ConnectionError("libusb could not read FreeSRP serial  
}  
else
```

[actcd19/main/s/supertux/supertux_0.6.0-1/src/gui/item_intfield.cpp:84](#)

```
*number = new_number;  
} catch (...) {  
    input = std::to_string(*number);  
}  
}
```

[actcd19/main/libc/libcereal/libcereal_1.2.2-3/include/cereal/cereal.hpp:679](#)

```
auto iter = itsSharedPointerMap.find( id );  
if(iter == itsSharedPointerMap.end())  
    throw Exception("Error while trying to deserialize a smart po
```

```
return iter->second;
```

[actcd19/main/b/botan/botan_2.9.0-2/src/tests/test_pkcs11_low_level.cpp:214](#)

```
else
{
    result.test_rc_fail(name, "return value should be: " + std::to_string(
        static_cast< uint32_t >(rv));
}
```

[actcd19/main/f/freeorion/freeorion_0.4.8-1/Empire/Diplomacy.cpp:29](#)

```
std::string retval;
retval += "Diplomatic message from : " + std::to_string(m_sender_empire) +
    " to: " + std::to_string(m_recipient_empire) + " about: " + m_message;
switch (m_type) {
case WAR_DECLARATION:      retval += "War Declaration";
```

[actcd19/main/a/android-platform-frameworks-native/android-platform-frameworks-native_8.1.0+r23-2/libs/hwc2on1adapter/HWC2On1Adapter.cpp:1391](#)

```
static std::string approximateFloatString(float f) {
    if (static_cast<int32_t>(f) == f) {
        return std::to_string(static_cast<int32_t>(f));
    }
    int32_t truncated = static_cast<int32_t>(f * 10);
```

[actcd19/main/r/rocksdb/rocksdb_5.17.2-3/table/data_block_hash_index_test.cc:182](#)

```
ASSERT_EQ(original_size, map_offset);
for (uint8_t i = 0; i < 100; i++) {
    std::string key("key" + std::to_string(i));
    uint8_t restart_point = i;
```

```
ASSERT_TRUE(
```

[actcd19/main/u/ultracopier/ultracopier_1.6.1.3-1/plugins/CopyEngine/Ultracopier/ScanFileOrFolder.cpp:198](#)

```
{
    stopped=false;
    ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice,"start the
    destination=resolveDestination(QString::fromStdString(destination)).
    if(stopIt)
```

[actcd19/main/d/dnsdist/dnsdist_1.3.3-3/test-dnsdistpacketcache_cc.cc:78](#)

```
size_t delcounter=0;
for(delcounter=0; delcounter < counter/1000; ++delcounter) {
    DNSName a=DNSName(std::to_string(delcounter))+DNSName(" hello");
    vector<uint8_t> query;
    DNSPacketWriter pwQ(query, a, QType::A, QClass::IN, 0);
```

[actcd19/contrib/o/openmw/openmw_0.45.0-2/components/esm/mappings.cpp:57](#)

```
        default:
            throw std::runtime_error("PartReferenceType " +
                std::to_string(type) + " not associated with a mesh"
        }
    }
```

[actcd19/main/t/thunderbird/thunderbird_60.5.1-1/ipc/glue/GeckoChildProcessHost.cpp:1025](#)

```
#endif // defined(MOZ_SANDBOX)
    mLaunchOptions->handles_to_inherit.push_back(reinterpret_cast<HANDLE>
    std::string hStr = std::to_string(h);
    cmdLine.AppendLooseValue(UTF8ToWide(hStr));
```

```
}
```

[actcd19/main/m/mysql-workbench/mysql-workbench_6.3.10+dfsg-3/.pc/build-i386-json.patch/library/forms/jsonview.cpp:432](#)

```
auto newNode = (updateMode) ? node : node->add_child();
generateTree(objectName.empty() ? jv : obj[objectName], 0, newNode);
newNode->set_string(0, objectName + "{" + std::to_string(obj[objectName].value()).
newNode->set_tag(objectName);
_dataChanged(false);
```

[actcd19/main/m/mysql-workbench/mysql-workbench_6.3.10+dfsg-3/backend/wbprivate/sqlide/wb_live_schema_tree.cpp:1520](#)

```
limit_rows = false;
if (limit_rows)
    caption += _(" - Limit ") + std::to_string(limit_rows_count);
}
view_item.caption = caption;
```

[actcd19/main/s/spades/spades_3.13.0+dfsg2-2/src/common/assembly_graph/core/graph_iterators.hpp:173](#)

```
Comparator(), bool canonical_only = false)
: SmartIterator<Graph, VertexId, Comparator>(
    g, "SmartVertexIterator " + std::to_string(get_id()), t,
    comparator, canonical_only) {
this->insert(g.begin(), g.end());
```

[actcd19/main/f/flang/flang_20181226-2/flang-driver/unittests/Tooling/ExecutionTest.cpp:271](#)

```

for (unsigned i = 1; i <= NumFiles; ++i) {
    std::string File = "f" + std::to_string(i) + ".cc";
    std::string Symbol = "loong_function_name_" + std::to_string(i);
    Files.push_back(File);
    FileToContent[File] = "void " + Symbol + "() {}";

```

[actcd19/main/t/toulbar2/toulbar2_1.0.0+dfsg3-2/src/utils/tb2reader.cpp:1513](#)

```

        exit(EXIT_FAILURE);
    }
    streamContentVec.push_back(std::make_pair(GCFTemplate[i], s
}
// ----- Read variable and add it to stream

```

[actcd19/main/c/chromium/chromium_72.0.3626.121-1/v8/src/torque/declarations.cc:278](#)

```

for (auto type : specialized_types) {
    std::string type_string = type->MangledName();
    result += std::to_string(type_string.size()) + type_string;
}
return result;

```

[actcd19/main/f/ffms2/ffms2_2.23-3/src/core/ffms.cpp:354](#)

```

ReplaceString(s, "%samplerate%", std::to_string(AP->SampleRate));
ReplaceString(s, "%channels%", std::to_string(AP->Channels));
ReplaceString(s, "%bps%", std::to_string(AP->BitsPerSample));
ReplaceString(s, "%delay%", std::to_string(static_cast<int>(AP->Fir

```

[actcd19/main/f/freeorion/freeorion_0.4.8-1/UI/TechTreeWnd.cpp:1704](#)

```

        cost_btn->SetText(cost_str + just_pad + just_pad);

        std::string time_str = std::to_string(this_row_tech->ResearchTime(c
        if (GG::Button* time_btn = dynamic_cast<GG::Button*>((size() >= 4)
            time_btn->SetText(time_str + just_pad + just_pad);

```

[actcd19/main/b/biboumi/biboumi_8.3-1/src/bridge/bridge.cpp:550](#)

```

        {
            auto& list = channel_list_cache[iid.get_server()];
            const auto res = this->send_matching_channel_list(list, rs_
            return res;
        }

```

[actcd19/main/u/ultracopier/ultracopier_1.6.1.3-1/plugins/Themes/Oxygen/ThemesFactory.cpp:449](#)

```

        currentSpeed=1024*64;
        if(optionsEngine!=NULL)
            optionsEngine->setOptionValue("currentSpeed",std::t
        else
            ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Cr

```

[actcd19/main/c/chromium/chromium_72.0.3626.121-1/ui/ozone/platform/wayland/wayland_buffer_manager.cc:253](#)

```

        if (planes_count != strides.size() || planes_count != offsets.size()
            planes_count != modifiers.size()) {
            reason = "Number of strides(" + std::to_string(strides.size()) +
                ")/offsets(" + std::to_string(offsets.size()) + ")/modifiers(" +
                std::to_string(modifiers.size()) +

```

[actcd19/main/p/psi4/psi4_1.2.1-2/psi4/src/psi4/libqt/timer.cc:603](#)

```
        str += key_;\n        str += " on thread ";\n        str += std::to_string(thread_rank);\n        str += " are on and cannot be merged.";\n        throw PsiException(str, __FILE__, __LINE__);\n    }\n};
```

[actcd19/main/libfilezilla/libfilezilla_0.15.1-1/lib/libfilezilla/string.hpp:194](#)

```
inline typename std::enable_if<std::is_arithmetic<std::decay_t<Arg>>::value>\n{\n    return std::to_string(std::forward<Arg>(arg));\n}
```

[actcd19/main/s/simgrid/simgrid_3.21+dfsg-4/examples/s4u/dht-chord/s4u-dht-chord-node.cpp:134](#)

```
try {\n    simgrid::s4u::Mailbox::by_name(std::to_string(pred_id_))->put(suc\n} catch (simgrid::TimeoutError& e) {\n    XBT_DEBUG("Timeout expired when sending a 'SUCCESSOR_LEAVING' to
```

[actcd19/main/g/gnss-sdr/gnss-sdr_0.0.10-4/src/tests/unit-tests/signal-processing-blocks/acquisition/galileo_e5a_pcps_acquisition_gsoc2014_gensource_test.cc:417](#)

```
config->set_property("Acquisition_5X.item_type", "gr_complex");\nconfig->set_property("Acquisition_5X.coherent_integration_time_ms",\n    std::to_string(integration_time_ms));\nconfig->set_property("Acquisition_5X.max_dwells", "1");\nconfig->set_property("Acquisition_5X.implementation", "Galileo_E5a_
```

[actcd19/main/o/ompl/ompl_1.4.2+ds1-1/demos/HypercubeBenchmark.cpp:110](#)

```
ompl::tools::Benchmark::Request request(runtime_limit, memory_limit);
ompl::tools::Benchmark b(ss, "HyperCube");
b.addExperimentParameter("num_dims", "INTEGER", std::to_string(ndim));

addPlanner(b, std::make_shared<ompl::geometric::STRIDE>(ss.getSpace()));
```

[actcd19/main/c/chromium/chromium_72.0.3626.121-1/third_party/dawn/src/common/DynamicLib.cpp:87](#)

```
if (proc == nullptr && error != nullptr) {
    *error = "Windows Error: " + std::to_string(GetLastError());
}
#elif DAWN_PLATFORM_POSIX
```

[actcd19/main/p/psi4/psi4_1.2.1-2/psi4/src/psi4/libmints/cdsalclist.cc:312](#)

```
name += com.coef > 0.0 ? "+" : "-";
name += std::to_string(std::fabs(com.coef)) + " ";
name += molecule_>label(com.atom);
if (com.xyz == 0)
```

[actcd19/main/u/ultracopier/ultracopier_1.6.1.3-1/plugins/CopyEngine/Ultracopier/TransferThread.cpp:767](#)

```
void TransferThread::tryCopyDirectly()
{
    ULTRACOPIER_DEBUGCONSOLE(Ultracopier::DebugLevel_Notice, "[" + std::to_string(id) + "]");

    sendState_readStopped = false;
```

[actcd19/main/w/widelands/widelands_19+repack-6/src/wui/plot_area.cc:438](#)

```
*/  
void WuiPlotArea::draw(RenderTarget& dst) {  
    draw_plot(dst, get_inner_h() - kSpaceBottom, std::to_string(highest  
}  

```

[actcd19/main/p/psi4/psi4_1.2.1-2/psi4/src/psi4/detci/params.cc:604](#)

```
if (i < 1 || i > Parameters_->num_roots) {  
    std::string str = "Invalid number of states to average (";  
    str += std::to_string(i);  
    str += ")";  
    throw PsiException(str, __FILE__, __LINE__);  
}
```

[actcd19/main/p/pdns-recursor/pdns-recursor_4.1.11-1/filterpo.cc:347](#)

```
if (addr.isIPv4()) {  
    const uint8_t* bytes = reinterpret_cast<const uint8_t*>(&addr.sin4.  
    res += DNSName(std::to_string(bytes[3]) + "." + std::to_string(byte  
}  
else {
```

[actcd19/main/t/trilinos/trilinos_12.12.1-7/packages/stk/stk_simd/unit_tests/SimdFixture.hpp:268](#)

```
virtual std::string get_input_vals(int i) const override {  
    return std::to_string(x.scalar(i))+" "+std::to_string(y.scalar(i));  
}  
};
```

[actcd19/main/p/performous/performous_1.1+git20181118-2/game/3dobject.cc:83](#)

```

        m_faces.push_back(f);
    } else {
        throw std::runtime_error("Invalid face in '");
    }
}

```

[actcd19/main/f/freeorion/freeorion_0.4.8-1/util/SitRepEntry.cpp:227](#)

```

        UserStringNop("SITREP_SHIP_DAMAGED_AT_SYSTEM_LABEL"), t
        sitrep.AddVariable(VarText::SHIP_ID_TAG,          std::to_string(c
        sitrep.AddVariable(VarText::DESIGN_ID_TAG,        std::to_string(s

    } else if (auto planet = std::dynamic_pointer_cast<const Planet>(ok

```

[actcd19/main/o/opensurgesim/opensurgesim_0.7.0-8/SurgSim/Devices/LabJack/UnitTests/LabJackDeviceTest.cpp:158](#)

```

    EXPECT_TRUE(consumer->m_lastReceivedInput.scalars().hasEntry(SurgSi
        std::to_string(0)); // The LabJackDevice provides entries
}

```

[actcd19/main/f/fw4spl/fw4spl_17.2.0-2/SrcLib/patch/fwMDSemanticPatch/src/fwMDSemanticPatch/V09ALA/V10/fwData/Image.cpp:109](#)

```

    if (label.empty())
    {
        label = "label_" + std::to_string(count++);
    }

```

§ References

§ Informative References

[CODESEARCH]

Andrew Tomazos. [Code search engine website](https://codesearch.isocpp.org). URL: <https://codesearch.isocpp.org>

[DRAGON]

Guy L. Steele Jr.; Jon L White. [How to Print Floating-Point Numbers Accurately](https://fmt.dev/papers/p372-steele.pdf). URL: <https://fmt.dev/papers/p372-steele.pdf>

[N1803]

Pete Becker. [Simple Numeric Access](https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2005/n1803.html). URL: <https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2005/n1803.html>

[N2408]

Pete Becker. [Simple Numeric Access Revision 2](https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2007/n2408.html). URL: <https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2007/n2408.html>

[N4910]

Thomas Köppe. [Working Draft, Standard for Programming Language C++](https://wg21.link/n4910). URL: <https://wg21.link/n4910>

[STD_DISCUSSION]

[Deprecate to_string\(\) for floating point types?](https://groups.google.com/a/isocpp.org/g/std-discussion/c/6xhWgsEuvPo). URL: <https://groups.google.com/a/isocpp.org/g/std-discussion/c/6xhWgsEuvPo>